



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AWS Certified AI Practitioner Training (AIF-C01)

TGS Ref No: TGS-2024049338

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	29 Sep 2024	First version	Tertiary Infotech Academy
2.0		New Format	Tertiary Infotech Academy
3.0	17 Oct 2025	Update company name	Tertiary Infotech Academy
4.0	20 Jul 2026	Realigned to official AIF-C01 exam domains; teaching order follows the deck (Domain 1–5) with contiguous slide ranges; Slides column populated from the built deck.	Tertiary Infotech Academy
5.0	21 Jul 2026	Reformatted to the current house lesson-plan layout (Course Information, Learning Outcomes, Learning Reinforcement, Course Schedule, Lab Reference); day-by-day schedule as Time / Duration / Topic-Activity with a Slides column retained; Lab Reference table maps each exam domain, its weighting and its labs. Deck and 2-day / 14-hour duration unchanged.	Tertiary Infotech Academy

TABLE OF CONTENTS

Course Information.....	4
Learning Outcomes.....	4
Learning Reinforcement.....	4
Course Schedule.....	5
Day 1 — Domains 1 & 2 (Fundamentals of AI/ML and Generative AI).....	5
Day 2 — Domains 3, 4 & 5 + Assessment.....	6
Lab Reference (aligned to the exam domains).....	8
Assessment.....	8

Course Information

Course Title	AWS Certified AI Practitioner (AIF-C01)
Course Code	TGS-2024049338
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days — 7 training hours per day (14 training hours + 2 assessment hours)
Daily Timing	9:00 AM – 6:00 PM (1-hour lunch; tea breaks within training). Day 2 runs to 6:40 PM to include the assessment.
Mode	Instructor-led, with demonstrations and hands-on labs in the learner's own AWS account
Prerequisites	Basic IT literacy; an AWS account with Amazon Bedrock model access requested
Tools	AWS Management Console, AWS CLI, Amazon Bedrock, SageMaker, Amazon Q Business, Amazon Macie, and the 25 course labs

Course Overview

This 2-day hands-on course prepares participants for the AWS Certified AI Practitioner (AIF-C01) certification. Through instructor-led sessions and 25 structured labs, participants gain practical skills across the five official exam domains: Fundamentals of AI and ML (20%); Fundamentals of Generative AI (24%); Applications of Foundation Models (28%); Guidelines for Responsible AI (14%); and Security, Compliance and Governance for AI Solutions (14%).

Learning Outcomes

- LO1 — Implement AI applications with Amazon Q Business and AWS AI models.
- LO2 — Deploy AWS AI Bedrock workflows according to plan and machine learning techniques.
- LO3 — Identify and report issues with AI applications, data and implementation procedures.
- LO4 — Maintain data interoperability during AI development per data-management principles.
- LO5 — Perform data cleaning techniques to optimise AWS foundation models.

Learning Reinforcement

Learning is reinforced continuously in class rather than through theory alone:

- Every concept is followed by a hands-on lab that applies it in the learner's own AWS account.
- Each lab ends with a verification step so learners confirm their own output before moving on.
- Lab outputs build toward the practical assessment — the same techniques are re-used in the case-study tasks (A1–A5).
- Labs that create billing resources include a cost warning and a cleanup step.
- Day 2 closes with the WSQ written and practical assessments and a recap against the learning outcomes.

Course Schedule

Day 1 — Domains 1 & 2 (Fundamentals of AI/ML and Generative AI)

Time	Duration	Topic / Activity	Slides
9:00 – 9:15	15 min	Registration & Digital Attendance (AM) — trainer/learner intro, course structure, AWS account & Bedrock model-access check	1–22
9:15 – 9:45	30 min	DOMAIN 1 — Fundamentals of AI and ML (20%) Lecture: AI vs ML vs deep learning; learning types; inferencing; data types	24–51
9:45 – 10:30	45 min	Lab 1: AI, ML and Deep Learning Terminology [Demo] Lab 2: Data Types and Labelling for ML [Demo] Guided walkthrough; full labs self-paced in own account	25–51
10:30 – 10:45	15 min	Tea Break	—
10:45 – 11:45	60 min	Lecture: AI use cases & AWS managed AI services; ML development lifecycle Lab 3: AWS Managed AI Services Tour [Demo]	52–135
11:45 – 12:15	30 min	Lab 4: Train and Deploy a Model with SageMaker [Self-paced] Lab 5: The ML Pipeline and Data Preparation [Self-paced] Overview + cost/cleanup briefing; run self-paced	84–135
12:15 – 1:15	60 min	Lunch Break	—
1:15 – 1:25	10 min	Digital Attendance (PM)	—
1:25 – 2:25	60 min	DOMAIN 2 — Fundamentals of Generative AI (24%) Lecture: foundation models; generative AI concepts; tokens & embeddings	136–164
2:25 – 3:10	45 min	Lab 6: Amazon Bedrock Getting Started [Demo] Lab 7: Comparing Foundation Models and Inference Parameters [Demo] Lab 8: Tokens, Embeddings and Vector Representations [Demo]	137–164

3:10 – 3:25	15 min	Tea Break	—
3:25 – 4:25	60 min	Lecture: capabilities vs limitations; AWS infra for GenAI; service selection Lab 9: Generative AI Capabilities and Limitations [Demo] Lab 10: Choosing Bedrock, SageMaker JumpStart or a Managed Service [Demo]	165–182
4:25 – 5:05	40 min	Lab 11: Amazon Q Business Getting Started [Self-paced] Overview; run self-paced in own account	173–182
5:05 – 5:50	45 min	Case Study — Generative AI adoption (Retail / Amazon Q Business)	137–182
5:50 – 6:00	10 min	Day 1 Review & Q&A	—

Day 2 — Domains 3, 4 & 5 + Assessment

Time	Duration	Topic / Activity	Slides
9:00 – 9:10	10 min	Digital Attendance (AM) — Day 1 recap	—
9:10 – 10:10	60 min	DOMAIN 3 — Applications of Foundation Models (28%) Lecture: design considerations; prompt engineering	183–235
10:10 – 10:55	45 min	Lab 12: Prompt Engineering Basics — Zero-Shot and Few-Shot [Demo] Lab 13: Advanced Prompting and Prompt Templates [Demo] Lab 14: Prompt Misuse, Injection and Defences [Demo]	210–235
10:55 – 11:10	15 min	Tea Break	—
11:10 – 12:10	60 min	Lecture: RAG; fine-tuning; model evaluation Lab 15: RAG with Knowledge Bases for Amazon Bedrock [Self-paced] Lab 16: Vector Stores and Retrieval Quality [Self-paced] Overview + cost/cleanup briefing; run self-paced	184–271

12:10 – 1:10	60 min	Lunch Break	—
1:10 – 1:20	10 min	Digital Attendance (PM)	—
1:20 – 1:50	30 min	Lab 17: Fine-Tuning and Model Customisation [Self-paced] Lab 18: Evaluating Foundation Model Performance [Self-paced] Overview; run self-paced (Lab 17 config only — do not provision)	236–247, 248–271
1:50 – 2:40	50 min	DOMAIN 4 — Guidelines for Responsible AI (14%) Lecture: responsible AI dimensions; transparency & explainability Lab 19: Amazon Bedrock Guardrails [Demo]	272–297
2:40 – 3:20	40 min	Lab 20: Bias Detection with SageMaker Clarify [Self-paced] Lab 21: Explainability and Feature Attribution [Demo] Lab 22: Model Cards and Transparency [Demo]	273–284, 285–297
3:20 – 3:35	15 min	Tea Break	—
3:35 – 4:25	50 min	DOMAIN 5 — Security, Compliance & Governance (14%) Lecture: securing AI; governance & compliance Lab 23: Securing AI Workloads with IAM and KMS [Self-paced] Lab 24: Data Protection with Amazon Macie [Self-paced] Lab 25: Governance, Audit and Model Lineage [Demo]	298–374
4:25 – 4:40	15 min	Assessment Briefing — WSQ assessment instructions and rules	284
4:40 – 5:40	60 min	WSQ Written Assessment — Short-Answer Questions (WA-SAQ), 11 questions	368–374
5:40 – 6:40	60 min	WSQ Practical Assessment (CS) — lab-based practical tasks A1–A5	—
6:40	—	Assessment submission & close	—

Lab Reference (aligned to the exam domains)

All 25 hands-on labs mirror the slide deck and Learner Guide. Labs tagged [Demo] are walked through in class; labs tagged [Self-paced] are completed in the learner's own AWS account (several create billing resources — always complete the cleanup step).

Exam Domain / Topic	Weighting	Labs	Slides
DOMAIN 1: Fundamentals of AI and ML	20%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5	25–51, 52–83, 84–135
DOMAIN 2: Fundamentals of Generative AI	24%	Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11	137–164, 165–172, 173–182
DOMAIN 3: Applications of Foundation Models	28%	Lab 12, Lab 13, Lab 14, Lab 15, Lab 16, Lab 17, Lab 18	184–209, 210–235, 236–247, 248–271
DOMAIN 4: Guidelines for Responsible AI	14%	Lab 19, Lab 20, Lab 21, Lab 22	273–284, 285–297
DOMAIN 5: Security, Compliance, and Governance for AI Solutions	14%	Lab 23, Lab 24, Lab 25	299–333, 334–374

Assessment

Participants are assessed by two WSQ instruments on Day 2 afternoon:

- Written Assessment — Short-Answer Questions (WA-SAQ): 11 open-ended knowledge questions (K1–K11), 1 hour, open book.
- Practical Assessment (Case Study): lab-based practical tasks mapped to abilities A1–A5, 1 hour, open book. Each task is completed using the course labs.

A minimum competency grade of Competent (C) in both instruments is required to satisfy WSQ requirements.

Total Training Hours: 14 hrs · Total Assessment Hours: 2 hrs